Practical No.: 6

Aim –

Use string functions and regex for email validation, data extraction, and text analysis.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

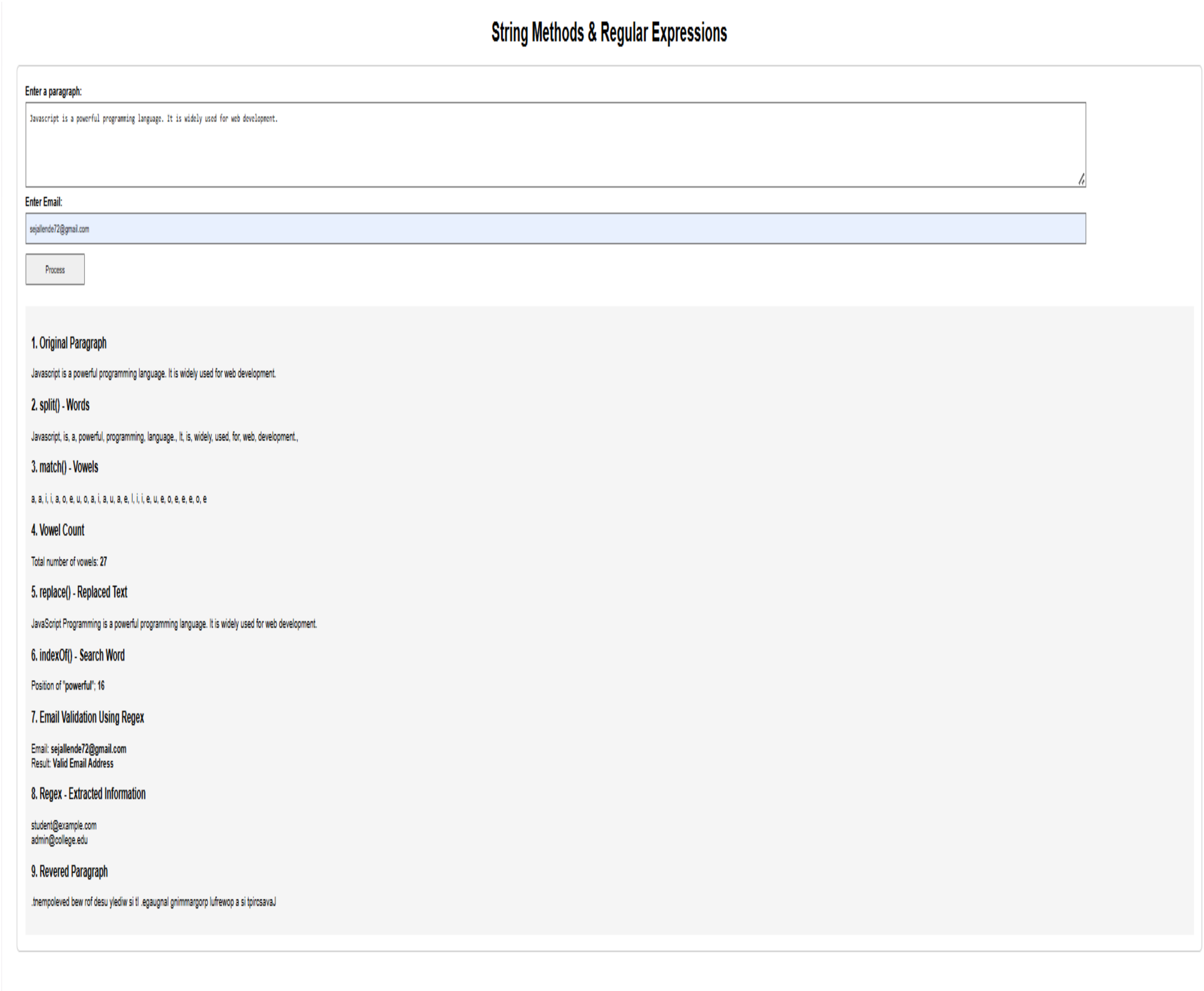
Experiment Program Code –

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      String Methods and Regular Expressions
    </title>

    <style>
      body {
        font-family: Arial, sans-serif;
        margin: 40px;
      }

      h1 {
        text-align: center;
      }

      .box {
        border: 1px solid #ccc;
        padding: 20px;
        margin-top: 20px;
        border-radius: 8px;
      }

      input, textarea, button {
        padding: 10px;
        margin: 5px 0;
        width: 90%;
      }

      button {
        width: 150px;
        cursor: pointer;
      }
```

```
        #output {
            margin-top: 20px;
            padding: 15px;
            background-color: #f5f5f5;
        }
    </style>
</head>

<body>
    <h1>String Methods & Regular Expressions</h1>
    <div class="box">
        <label><b>Enter a paragraph:</b></label> <br>

        <textarea id="paragraph" rows="5" >Javascript is a powerful programming language. It is widely used for web development.
        </textarea>
        <br>

        <label><b>Enter Email:</b></label> <br>
        <input type="text" id="email" placeholder="Enter your email"> <br>

        <button onclick="processString()">Process</button>
        <div id="output"></div>
    </div>

    <script>
        function processString() {
            let paragraph = document.getElementById("paragraph").value;
            let email = document.getElementById("email").value;

            let words = paragraph.split(/\s+/);

            let vowels = paragraph.match(/[aeiou]/gi);
            let vowelCount = vowels ? vowels.length : 0;

            let replacedParagraph = paragraph.replace(
                /JavaScript/gi,
                "JavaScript Programming"
            );

            let searchWord = "powerful";
            let position = paragraph.indexOf(searchWord);

            let emailRegex =
            /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;

            let emailResult;

            if(emailRegex.test(email)) {
                emailResult = "Valid Email Address";
            } else {
                emailResult = "Invalid Email Address";
            }

            let emailText =
            "For queries contact student@example.com or admin@college.edu";

            let extractedEmails = emailText.match(
                /[a-zA-Z0-9._%+-]+@[a-zA-z0-9.-]+\.[a-zA-Z]{2,}/g
            );

            let reeversedParagraph = paragraph.split("").reverse().join("");

            document.getElementById("output").innerHTML = `
            <h3>1. Original Paragraph</h3>
            <p>${paragraph}</p>

            <h3>2. split() - Words</h3>
            <p>${words.join(", ")} </p>

            <h3>3. match() - Vowels</h3>
            <p>${vowels ? vowels.join(", ") : "No vowels found"}</p>

            <h3>4. Vowel Count</h3>
```

```
            <p>Total number of vowels: <b>${vowelCount}</b></p>

            <h3>5. replace() - Replaced Text</h3>
            <p>${replacedParagraph}</p>

            <h3>6. indexOf() - Search Word</h3>
            <p> Position of "<b>${searchWord}</b>";
            <b>${position}</b>
            </p>

            <h3>7. Email Validation Using Regex</h3>
            <p>
                Email: <b>${email}</b><br>
                Result: <b>${emailResult}</b>
            </p>

            <h3>8. Regex - Extracted Information</h3>
            <p>
                ${extractedEmails ? extractedEmails.join("<br>") : "No email address found"}
            </p>

            <h3>9. Revered Paragraph</h3>
            <p>${reeversedParagraph}</p>
            `;

        }
    </script>
  </body>
</html>
```

Output –


String Methods & Regular Expressions
Enter a paragraph:
Javascript is a powerful programming language. It is widely used for web development.
Enter Email:
sejallende72@gmail.com
Process
1. Original Paragraph
Javascript is a powerful programming language. It is widely used for web development.
2. split() - Words
Javascript, is, a, powerful, programming, language., It, is, widely, used, for, web, development.,
3. match() - Vowels
a, a, i, i, a, o, e, u, o, a, i, a, u, a, e, I, i, i, e, u, e, o, e, e, e, o, e
4. Vowel Count
Total number of vowels: 27
5. replace() - Replaced Text
JavaScript Programming is a powerful programming language. It is widely used for web development.
6. indexOf() - Search Word
Position of "powerful"; 16
7. Email Validation Using Regex
Email: sejallende72@gmail.com
Result: Valid Email Address
8. Regex - Extracted Information
student@example.com
admin@college.edu
9. Revered Paragraph
.tnempoleved bew rof desu ylediw si tI .egaugnal gnimmargorp lufrewop a si tpircsavaJ

Case Study Title –

To reverse a string and count the number of vowels in a given string or paragraph.

Case Study Program Code –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>String Programs | Sejal Lende</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    }

    body {
      min-height: 100vh;
      background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
      display: flex;
      justify-content: center;
      align-items: center;
      padding: 20px;
    }

    .container {
      background: rgba(255, 255, 255, 0.95);
      border-radius: 20px;
      box-shadow: 0 20px 40px rgba(0, 0, 0, 0.2);
      width: 100%;
      max-width: 700px;
      padding: 30px;
      overflow: hidden;
    }

    .header {
      text-align: center;
      margin-bottom: 30px;
      padding-bottom: 20px;
      border-bottom: 3px solid #667eea;
    }

    .header h1 {
      color: #333;
      font-size: 28px;
      margin-bottom: 8px;
    }

    .header .student-info {
      background: linear-gradient(90deg, #667eea, #764ba2);
      color: white;
      display: inline-block;
      padding: 8px 20px;
```

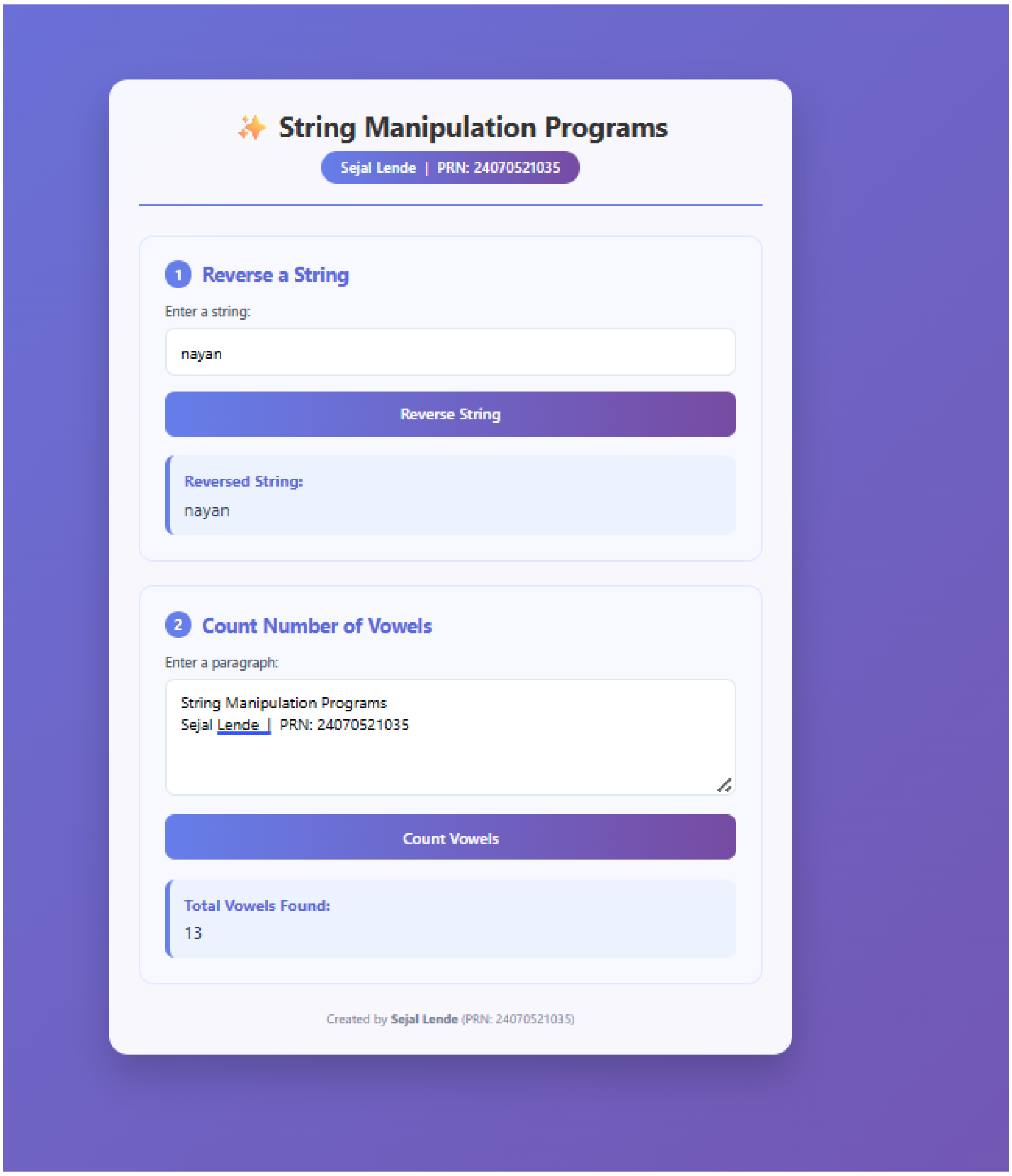
```css
    border-radius: 50px;
    font-size: 14px;
    font-weight: 600;
    letter-spacing: 0.5px;
}

.program-card {
    background: #f8f9ff;
    border-radius: 15px;
    padding: 25px;
    margin-bottom: 25px;
    border: 1px solid #e0e7ff;
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.program-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 10px 25px rgba(102, 126, 234, 0.2);
}

.program-card h2 {
    color: #5a67d8;
    font-size: 20px;
    margin-bottom: 15px;
    display: flex;
    align-items: center;
    gap: 10px;
}

.program-card h2 span {
    background: #667eea;
    color: white;
    width: 28px;
    height: 28px;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 14px;
}

label {
    display: block;
    font-weight: 600;
    color: #4a5568;
    margin-bottom: 8px;
    font-size: 14px;
}

input[type="text"],
textarea {
    width: 100%;
    padding: 12px 15px;
    border: 2px solid #e2e8f0;
    border-radius: 10px;
    font-size: 16px;
    transition: border-color 0.3s;
    outline: none;
    background: white;
}

input[type="text"]:focus,
textarea:focus {
    border-color: #667eea;
}
```

```
textarea {
  min-height: 120px;
  resize: vertical;
}

.btn {
  background: linear-gradient(90deg, #667eea, #764ba2);
  color: white;
  border: none;
  padding: 12px 28px;
  border-radius: 10px;
  font-size: 16px;
  font-weight: 600;
  cursor: pointer;
  margin-top: 15px;
  transition: transform 0.2s, box-shadow 0.2s;
  width: 100%;
}

.btn:hover {
  transform: scale(1.02);
  box-shadow: 0 5px 15px rgba(102, 126, 234, 0.4);
}

.btn:active {
  transform: scale(0.98);
}

.result {
  margin-top: 20px;
  padding: 15px;
  background: #edf2ff;
  border-radius: 10px;
  border-left: 5px solid #667eea;
  display: none;
}

.result.show {
  display: block;
  animation: fadeIn 0.4s ease;
}

.result strong {
  color: #5a67d8;
}

.result p {
  margin-top: 5px;
  font-size: 18px;
  color: #2d3748;
  word-break: break-word;
}

.footer {
  text-align: center;
  margin-top: 10px;
  color: #718096;
  font-size: 13px;
}

@keyframes fadeIn {
  from { opacity: 0; transform: translateY(10px); }
  to { opacity: 1; transform: translateY(0); }
}
```

```
    </style>
</head>
<body>
    <div class="container">
        <!-- Header with Student Details -->
        <div class="header">
            <h1>✨ String Manipulation Programs</h1>
            <div class="student-info">
                Sejal Lende  |  PRN: 24070521035
            </div>
        </div>

        <!-- Program 1: Reverse a String -->
        <div class="program-card">
            <h2><span>1</span> Reverse a String</h2>
            <label for="stringInput">Enter a string:</label>
            <input type="text" id="stringInput" placeholder="Type something here...">
            <button class="btn" onclick="reverseString()">Reverse String</button>

            <div class="result" id="reverseResult">
                <strong>Reversed String:</strong>
                <p id="reversedText"></p>
            </div>
        </div>

        <!-- Program 2: Count Vowels -->
        <div class="program-card">
            <h2><span>2</span> Count Number of Vowels</h2>
            <label for="paragraphInput">Enter a paragraph:</label>
            <textarea id="paragraphInput" placeholder="Write or paste a paragraph here..."></textarea>
            <button class="btn" onclick="countVowels()">Count Vowels</button>

            <div class="result" id="vowelResult">
                <strong>Total Vowels Found:</strong>
                <p id="vowelCount"></p>
            </div>
        </div>

        <div class="footer">
            Created by <strong>Sejal Lende</strong> (PRN: 24070521035)
        </div>
    </div>

    <script>
        // Program 1: Reverse a String
        function reverseString() {
            const input = document.getElementById("stringInput").value;
            const reversed = input.split("").reverse().join("");

            document.getElementById("reversedText").textContent = reversed || "(empty string)";
            document.getElementById("reverseResult").classList.add("show");
        }

        // Program 2: Count number of vowels in a paragraph
        function countVowels() {
            const text = document.getElementById("paragraphInput").value;
            const vowels = "aeiouAEIOU";
            let count = 0;

            for (let i = 0; i < text.length; i++) {
                if (vowels.includes(text[i])) {
                    count++;
                }
            }
```

```
            document.getElementById("vowelCount").textContent = count;
            document.getElementById("vowelResult").classList.add("show");
        }

        // Optional: Allow Enter key to trigger reverse
        document.getElementById("stringInput").addEventListener("keypress", function(e) {
            if (e.key === "Enter") reverseString();
        });
    </script>
</body>
</html>
```

Output –


✨ String Manipulation Programs
Sejal Lende | PRN: 24070521035
1 Reverse a String
Enter a string:
nayan
Reverse String
Reversed String:
nayan
2 Count Number of Vowels
Enter a paragraph:
String Manipulation Programs
Sejal Lende | PRN: 24070521035
Count Vowels
Total Vowels Found:
13
Created by Sejal Lende (PRN: 24070521035)

Conclusion –

The experiment was successfully performed to study various JavaScript string functions and regular expressions. Different string operations such as split(), match(), replace(), and indexOf() were used for text analysis. Regular expressions were also implemented for email validation and extraction of email addresses.

In the case study, a string was successfully reversed and the number of vowels and total characters were calculated. Thus, the practical helped in understanding string manipulation and the application of regular expressions in JavaScript.